Selection on Observables

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Overview

- This lecture covers approaches to estimating **causal effects** in settings with selection on observables
- Also known as: unconfoundedness, exogeneity, ignorability, conditional independence

Controlling Selection Bias: CIA

- A classic approach in the program evaluation literature is to adjust observational comparisons for differences in observed covariates
- This approach is motivated by a conditional independence assumption (CIA):

$$(Y_i(1), Y_i(0)) \perp T_i | X_i$$

- CIA is also called "unconfoundedness," "selection on observables," "exogeneity," "ignorability"

Controlling Selection Bias: CIA

$$(Y_i(1), Y_i(0)) \perp T_i | X_i$$

- The idea is that while treatment is not unconditionally random, it may be as good as random conditional on covariates
- Controlling for X_i then captures any systematic relationship between potential outcomes and treatment, eliminating selection bias
- This is the case in stratified RCTs and may be true in observational data with the right controls

Controlling Selection Bias: CIA

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- The idea is that while treatment is not unconditionally random, it may be as good as random conditional on covariates
- Controlling for X_i then captures any systematic relationship between potential outcomes and treatment, eliminating selection bias
- This is the case in stratified RCTs and may be true in observational data with the right controls
- Is this testable? Usually not but sometimes one can be "smart" and provide evidence in favor of CIA (e.g. Card-Heining-Kline, 2013) .

Full Covariate Matching

- Under CIA an obvious approach is to simply compare treatment and control groups conditional on the covariates:

$$\Delta(X_i) = E\left[Y_i | T_i = 1, X_i\right] - E\left[Y_i | T_i = 0, X_i\right]$$

- CIA implies

$$\begin{aligned}\Delta(X_i) &= E\left[Y_i(1) | T_i = 1, X_i\right] - E\left[Y_i(0) | T_i = 0, X_i\right] \\ &= E\left[Y_i(1) - Y_i(0) | X_i\right] \\ &\equiv ATE(X_i).\end{aligned}$$

- Covariate-specific treatment/control contrasts capture conditional average treatment effects
- By computing $\Delta(X_i)$ for every value of X_i and then weighting appropriately, we can obtain any causal effect of interest

Full Covariate Matching

$$ATE = \sum_x Pr\left[X_i = x\right] \Delta(x)$$

$$TOT = \sum_x Pr\left[X_i = x | T_i = 1\right] \Delta(x)$$

$$TNT = \sum_x Pr\left[X_i = x | T_i = 0\right] \Delta(x)$$

OLS as Matching

$$Y_i = D_i'\Pi + \delta T_i + U_i$$

- Recall our discussion of OLS with a saturated set of dummies for all values of X_i. We showed

$$\delta = E\left[\frac{\sigma_T^2(X_i)}{E\left[\sigma_T^2(X_i)\right]} \cdot \Delta(X_i)\right]$$

see slide 42 of Lecture2

- OLS with saturated controls is a version of full covariate matching that generates a variance-weighted average treatment effect
- OLS and matching are not qualitatively different
  - Both are methods of adjusting for observables
  - Coincide when the controls are flexible enough

Side Note: Controls in OLS

- Suppose we are interested in estimating the structural equation:

$$Y_i = \alpha + \delta T_i + \epsilon_i$$

- We believe CIA holds, so we add controls for X_i:

$$Y_i = \alpha + \delta T_i + X_i' \gamma + e_i$$

- How should we interpret γ? What if we get a "weird" coefficient on γ?

Side Note: Controls in OLS

$$Y_i = \alpha + \delta T_i + \epsilon_i$$

- We are only trying to estimate the causal effect of T_i, not of X_i
- Consider the linear projection of the structural error ϵ_i on X_i:

$$\epsilon_i = X_i'\gamma + e_i$$

- This is an OLS regression, so $Cov(X_i, e_i) = 0$ by definition

Worker Bee Controls

- Substituting this projection into the structural equation yields

$$Y_i = \alpha + \delta T_i + X_i' \gamma + e_i$$

- The X's in this equation are "worker bee" controls meant to eliminate selection bias
- We are not trying to estimate causal effects of X_i so it doesn't make sense to worry that $Cov(X_i, e_i) \neq 0$
- It's important to be clear about which parameters are structural and which are projection coefficients on conditioning variables

Section 1

Matching

Matching with High-dimensional Covariates

- Full covariate matching may not be feasible if there are many values of X_i (the "curse of dimensionality")
- With imperfect overlap in the treatment and control covariate distributions, $\Delta(X_i)$ can't be computed for some values of X_i
- With continuous covariates every value of X_i is unique
- Matching with high-dimensional covariates requires comparing observations with "similar" values of X_i

The Propensity Score Theorem

- The propensity score theorem provides a path for reducing the dimensionality of the covariates
- The **propensity score** is the conditional probability of treatment, $e(X_i) = Pr[T_i = 1|X_i]$
- Rosenbaum and Rubin (1983) show that

$$(Y_i(1), Y_i(0)) \perp T_i|X_i \implies (Y_i(1), Y_i(0)) \perp T_i|e(X_i)$$

- Under CIA, it isn't necessary to fully condition on X_i to eliminate selection bias; it's sufficient to condition on the scalar $e(X_i)$
- There may be few treatment/control observations sharing values of X_i but many sharing values of $e(X_i)$

Proof of the Propensity Score Theorem

- Proof:

$$Pr\left[T_i = 1 | Y_i(1), Y_i(0), e(X_i)\right] = E\left[T_i | Y_i(1), Y_i(0), e(X_i)\right]$$

$$= E\left[E\left[T_i | Y_i(1), Y_i(0), X_i\right] | Y_i(1), Y_i(0), e(X_i)\right]$$

$$= E\left[E\left[T_i | X_i\right] | Y_i(1), Y_i(0), e(X_i)\right]$$

$$= E\left[e(X_i) | Y_i(1), Y_i(0), e(X_i)\right]$$

$$= e(X_i).$$

Balacing Distributions vs. Matching

- The p-score theorem says that we need to condition on the (potentially big) vector X_i but just on the propensity score (one dimensional object)
- $e(X_i)$ turns out to be the coarsest "balancing score" that yields full independence of treatments from potential outcomes given CIA assumption
- But how can it make sense to compare units with different X_i?
- Answer: it doesn't.
  - Difference of means is comparing distributions not units
  - As we get more data on each $e(X_i)$ cell, there are more matches with exactly the same X_i
    - But no need to stratify on X_i in estimation

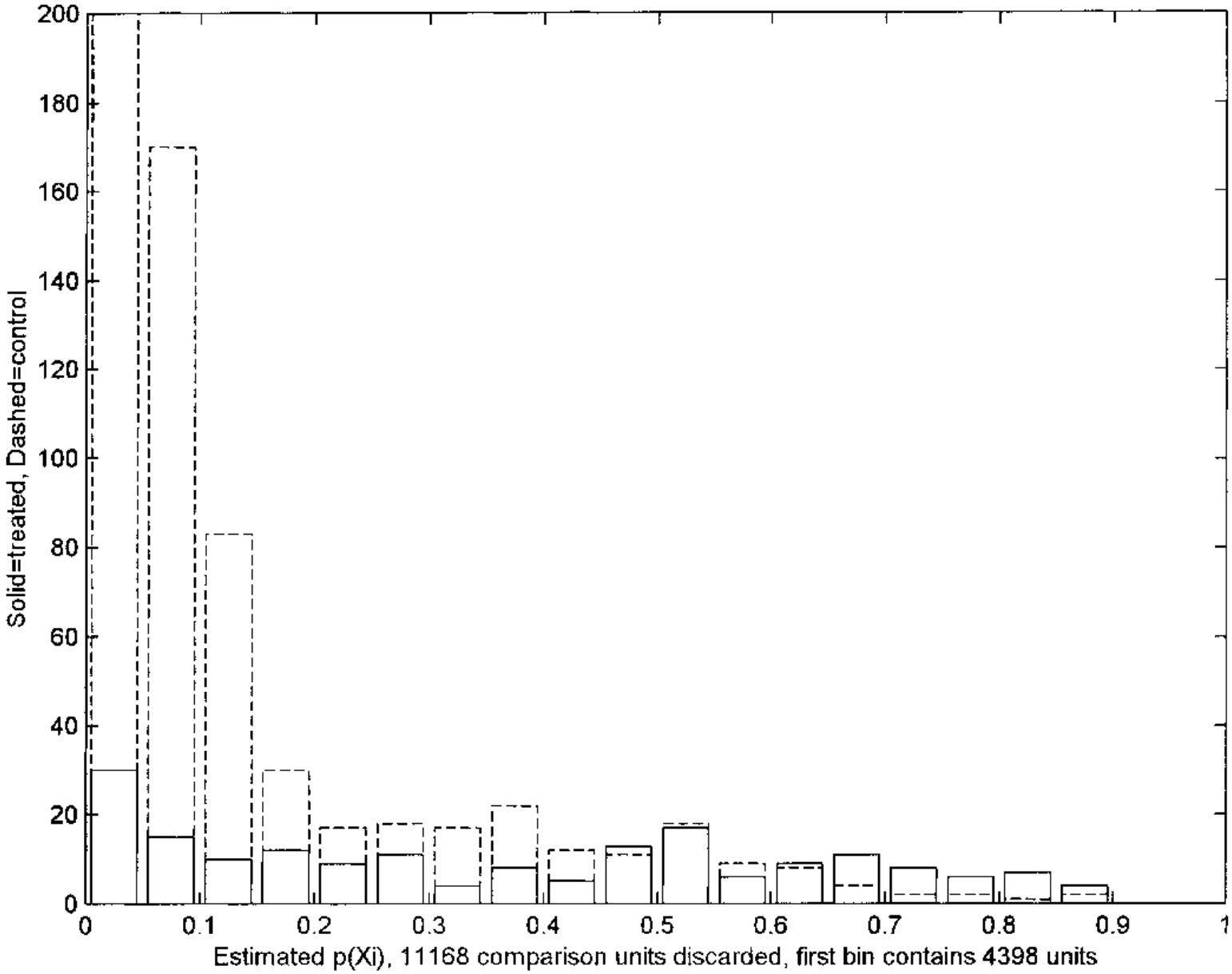
Estimating the Propensity Score

- The propensity score theorem shows that we can reduce the dimensionality of the covariates by matching on $e(X_i)$ rather than X_i
- But $e(X_i)$ is unknown except in special situations, so usually it must be estimated
- With discrete data $e(X_i)$ can be estimated using covariate-specific empirical frequencies (but then why not do full matching?)
- In practice the propensity score is often estimated with a flexible logit or probit
- Logit with 2nd-3rd degree polynomials in the continuous variables seem to be the way to go (MHE).

Overlap

- After estimating the propensity score it is common to assess the overlap between treatment and control distributions of $e(X_i)$
- Identification of the TOT requires $e(X_i) < 1 \; \forall X_i$
  - Requires constructing $E\left[Y_i(0) | T_i = 1\right]$, and there is no hope of imputing counterfactual outcomes for treated observations with $e(X_i) = 1$.
- Likewise, identifying the TNT requires $e(X_i) > 0$ and the ATE requires $0 < e(X_i) < 1$
- Crump et al. (2009): "trim" observations with small or large values of $e(X_i)$ to improve overlap and reduce variance
  - Note: this changes the estimand - but may be acceptable if effect heterogeneity is not of primary interest
  - Huber, Lechner, and Wunsch (2013): let fraction trimmed decline asymptotically to 0

FIGURE 1.—HISTOGRAM OF ESTIMATED PROPENSITY SCORE, NSW AND CPS



Nearest Neighbor Matching

- A further question is how to select observations with similar values of the propensity score
- One common approach is nearest neighbor matching
- Idea:

$$\hat{\delta}_{NN} = \left(\frac{1}{\sum_i T_i}\right) \sum_i T_i \left(Y_i - Y_{n(i)}\right),$$

$$n(i) = \arg \min_{j: T_j = 0} \left|\hat{e}(X_i) - \hat{e}(X_j)\right|$$

- The NN estimator compares each treated observation to the control observation with the closest value of the propensity score
- What are we estimating with $\hat{\delta}_{NN}$? ATE? TOT? TNT?

Problems with Nearest Neighbor Matching

- One problem with nearest neighbor matching is a lack of efficiency: $\hat{\delta}_{NN}$ uses only the control match with the closest value of $e(X_j)$ and ignores any other "close" matches
- A second problem is that it's difficult to get standard errors for nearest neighbor matching estimators
  - Abadie and Imbens (2008): bootstrap fails for nearest neighbor matching due to the non-smoothness of the nearest neighbor function
  - Abadie and Imbens (2012): standard errors based on martingale approximation

Kernel Matching

- Kernel matching estimators use all control observations to impute counterfactuals rather than just the nearest neighbor
- A kernel matching estimate of the TOT is given by

$$\hat{\delta}_K = \left(\frac{1}{\sum_i T_i}\right) \sum_i T_i \left[Y_i - \hat{\mu}_0 \left(\hat{e}(X_i)\right)\right],$$

$$\hat{\mu}_0(e) = \frac{\sum_j (1 - T_j) K\left(\frac{\hat{e}(X_j) - e}{h}\right) Y_j}{\sum_j (1 - T_j) K\left(\frac{\hat{e}(X_j) - e}{h}\right)}.$$

- $K(\cdot)$ is a symmetric kernel weighting function maximized at 0 and h is a bandwidth
- The kernel matching estimator compares each treated observation to a weighted average of control observations, with weights that are decreasing in the absolute value of the difference in propensity scores

Problems with Kernel Matching

- Kernel matching estimators have lower variance than NN estimators since they use more data, but also have more bias since they use information from more dissimilar matches
- As with NN estimators, there are some thorny implementation issues with kernel matching
  - Computing standard errors is complicated by the two-step nature of the estimation process (propensity score, then matching). This makes bootstrapping attractive but the validity of the bootstrap depends on the kernel
  - Kernel matching also requires choosing the bandwidth h. There are cross-validation choices available for bandwidth choice but these can behave poorly in finite samples

Section 2

Reweighting

Why matching? Can't we just re-weight?

Why matching? Can't we just re-weight?

- Propensity score reweighting sidesteps the matching step and reweights covariate distributions to match in the treatment and control groups
- Suppose we want the TOT. $E\left[Y_i(1)|T_i=1\right]$ is identified by the mean outcome in the treatment group; the hard part is identifying $E\left[Y_i(0)|T_i=1\right]$.
- Think about reweighting the control covariate distribution to match the treatment distribution. Defining $g_{X|T=t}(x)$ as the density (or mass) function of X_i conditional on $T_i=t$, this requires a set of weights $\omega(x)$ satisfying

$$\omega(x)g_{X|T=0}(x) = g_{X|T=1}(x) \ \forall x$$

Propensity Score Reweighting

- By Bayes rule we can write

$$g_{X|T=t}(x) = \frac{Pr\left[T_i = t|X_i = x\right] g_X(x)}{Pr\left[T_i = t\right]}$$

- The marginal density $g_X(x)$ cancels when we divide the conditional densities, yielding

$$\omega(x) = \frac{Pr\left[T_i = 1|X_i = x\right]}{Pr\left[T_i = 0|X_i = x\right]} \cdot \frac{Pr\left[T_i = 0\right]}{Pr\left[T_i = 1\right]}$$

$$= \frac{e(x)}{1 - e(x)} \cdot \frac{1 - Pr\left[T_i = 1\right]}{Pr\left[T_i = 1\right]}$$

Propensity Score Reweighting

- Now reweight the observed outcome distribution for the controls:

$$E\left[\omega(X_i)Y_i|T_i=0\right]=\int\int\omega(x)\cdot y\cdot f_{Y(0),X|T=0}(x,y)dydx$$

$$=\int\left[\int yf_{Y(0)|X,T=0}(y)dy\right]\omega(x)g_{X|T=0}(x)dx$$

$$=\int E\left[Y_i(0)|T_i=0,X_i=x\right]g_{X|T=1}(x)dx$$

$$=\int E\left[Y_i(0)|X_i=x\right]g_{X|T=1}(x)dx$$

$$=E\left[Y_i(0)|T_i=1\right]$$

Propensity Score Reweighting

$$E\left[\omega(X_i)\,Y_i \middle| T_i = 0\right] = E\left[Y_i(0) \middle| T_i = 1\right]$$

- The counterfactual mean for the treatment group can therefore be obtained by reweighting the control covariate distribution to match the treated distribution
- But why stopping at the mean?

Propensity Score Reweighting

$$E\left[\omega(X_i)\,Y_i \middle| T_i = 0\right] = E\left[Y_i(0) \middle| T_i = 1\right]$$

- The counterfactual mean for the treatment group can therefore be obtained by reweighting the control covariate distribution to match the treated distribution
- But why stopping at the mean?
- This can be done for any function $h(Y_i)$ so we can recover the entire distribution of $Y_i(0)$ for the treatment group, not just the mean!
- Can be used to answer questions like: what would the distribution of wages look like now if we had the unionionization rate observed in of the 1970's? See Di Nardo, Fortin and Lemieux (1996).

Propensity Score Reweighting

- Similarly, define

$$\tilde{\omega}_1(x) = \frac{Pr[T_i = 1]}{e(x)}, \tilde{\omega}_0(x) = \frac{Pr[T_i = 0]}{1 - e(x)}$$

- Then

$$E[\tilde{\omega}_1(X_i) Y_i | T_i = 1] = E[Y_i(1)]$$

$$E[\tilde{\omega}_0(X_i) Y_i | T_i = 0] = E[Y_i(0)]$$

- Weighting observations by the inverse of the conditional probability of their treatment assignments recovers population mean potential outcomes
  - Horvitz and Thompson (1952) estimator for dealing with unequal sampling probabilities

- Define $\tilde{Y}_i = Y_i (\tilde{\omega}_1(X_i) T_i + \tilde{\omega}_0(X_i)(1 - T_i))$. Then a regression of $\tilde{Y}_i$ on T_i yields the ATE.

Propensity Score Reweighting: Practical Issues

- Reweighting estimators are usually easier to implement than matching, and it's easier to obtain standard errors – bootstrapping works as long as the overlap condition is satisfied
- Reweighting performs much better when the model used to fit $e(X_i)$ is a good approximation. It is therefore important to conduct thorough specification testing after estimating the propensity score, e.g. by:
  - Comparing model fits to nonparametric treatment probabilities in covariate cells
  - Checking that reweighting balances the treatment and control distributions of key individual covariates
- These tests are based on the fact that $T_i \perp X_i | e(X_i)$. Covariate distributions cannot differ between treatment and control groups conditional on the true propensity score

Section 3

Decomposition

Decomposition Methods

- Matching and reweighting estimators are closely related to decomposition methods
- Decomposition methods are traditionally used to separate differences between groups into a component explained by observables and an unexplained component
- This is conceptually similar to decomposing a treatment/control difference into a component explained by controls (bias) and an unexplained component (treatment effect)

Oaxaca-Blinder Decompositions

- Standard tool in labor economics: the Oaxaca (1973)-Blinder (1973) decomposition
- Let's briefly review how OB decompositions work
- There are two groups, M's and W's
- Average outcomes are $\bar{Y}_M$ and $\bar{Y}_W$, where $\bar{Y}_g = E\left[Y_{ig} | group_i = g\right]$
- We hope to explain group differences with an observed covariate vector X_i

Oaxaca-Blinder Decompositions

- Quantity to be explained:

$$\Delta \equiv \bar{Y}_M - \bar{Y}_W$$

- Suppose we run a regression for each group

$$Y_{ig} = X'_{ig}\beta_g + \epsilon_{ig}$$

- X_{ig} includes a constant
- OLS coefficient vector $\beta_g = E\left[X_{ig}X'_{ig}|g\right]^{-1} E\left[X_{ig}Y_{ig}|g\right]$

Oaxaca-Blinder Decompositions

- By construction OLS fits group means:

$$\bar{Y}_g = \bar{X}_g' \beta_g.$$

- Therefore we can write

$$\Delta = \bar{X}_M' \beta_M - \bar{X}_W' \beta_W$$

$$= \left(\bar{X}_M - \bar{X}_W\right)' \beta_M + \bar{X}_W' \left(\beta_M - \beta_W\right)$$

- The OB decomposition splits Δ into a component explained by X's, and a component explained by β's
- First term answers the question: How much more would M's make than W's if both groups were paid like M's for observables?
- Second term answers the question: How much of Δ is due to differences in earnings within groups?

Oaxaca-Blinder Decompositions

$$\Delta = (\bar{X}_M - \bar{X}_W)' \beta_M + \bar{X}'_W (\beta_M - \beta_W)$$

- Can also write the alternative decomposition:

$$\Delta = (\bar{X}_M - \bar{X}_W)' \beta_W + \bar{X}'_M (\beta_M - \beta_W)$$

- New first term answers the question: How much more would M's make than W's if both groups were paid like W's for observables?
- Second term measures much of Δ is explained by the β's, weighting with M's characteristics

Section 4

How to Choose Controls?

Cookbook

- Need a model for how is treatment determined.
- Good institutional knowledge of the context is often key here.
- Nothing is definite in empirical work, the question is whether your proposed CIA research design is an improvement relative to existing literature.
- Still, how can one can come up with a reasonable research design based on CIA? Can this be outsourced to AI?
- We'll examine these big questions in class via an example based on Dale and Kruger (2002)

Dale and Krueger (2002)

ESTIMATING THE PAYOFF TO ATTENDING A MORE SELECTIVE COLLEGE: AN APPLICATION OF SELECTION ON OBSERVABLES AND UNOBSERVABLES*

STACY BERG DALE AND ALAN B. KRUEGER

Estimates of the effect of college selectivity on earnings may be biased because elite colleges admit students, in part, based on characteristics that are related to future earnings. We matched students who applied to, and were accepted by, similar colleges to try to eliminate this bias. Using the College and Beyond data set and National Longitudinal Survey of the High School Class of 1972, we find that students who attended more selective colleges earned about the same as students of seemingly comparable ability who attended less selective schools. Children from low-income families, however, earned more if they attended selective colleges.

- Suppose we have this simple model of potential outcomes where treatment (D_i) indicates whether individual i attended a more selective college (just to fix ideas, think of this as being an Ivy league or private university, e.g. Harvard).

$$Y_{i0} = \alpha + \eta_i \tag{1}$$

$$Y_{i1} = Y_{i0} + \rho \tag{2}$$

- Observed earnings:

$$Y_i = D_i Y_{i1} + (1 - D_i) Y_{i0} = \alpha + \rho D_i + \eta_i \tag{3}$$

- Running a regression of Y_i on D_i won't allow me to identify ρ!
- Let's introduce a CIA assumption. Suppose we have a set of variables, called that A_i, measured prior-to-treatment such that

$$\mathrm{E}[\eta_i | D_i, A_i] = \mathrm{E}[\eta_i | A_i] \tag{4}$$

- Further assume that

$$\eta_i = \gamma' A_i + \varepsilon_i \tag{5}$$

where $\mathrm{E}[\varepsilon_i | A_i] = 0$. This leads to

$$\mathrm{E}[Y_i | D_i, A_i] = \alpha + \rho D_i + \mathrm{E}[\eta_i | A_i] = \alpha + \rho D_i + \gamma' A_i \tag{6}$$

Dale and Kruger Research Design

- Ok great, but what is a reasonable choice of A_i?
- Dale and Kruger (2002): an unrestricted set of dummy variables capturing the set of colleges that students applied to and got admitted to.
- Once controlling for this, the decision to attend a more "elite" university does not *systematically* depend on ε_i.

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- DK Research design thus compares differences in earnings among students that applied and got admitted to the same set of schools (e.g. Duke and North Carolina) but where one decided to go to North Carolina and the other went to Duke.

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- DK Research design thus compares differences in earnings among students that applied and got admitted to the same set of schools (e.g. Duke and North Carolina) but where one decided to go to North Carolina and the other went to Duke.
- Regression run by DK

$$Y_i = \alpha + \rho D_i + \underbrace{\delta_0 C_i + \sum_{j=1}^{150} \delta_j Group_j}_{\gamma' A_i} + \varepsilon_i \tag{7}$$

TABLE I

ILLUSTRATION OF HOW MATCHED-APPLICANT GROUPS WERE CONSTRUCTED

		Student applications to college							
		Application 1		Application 2		Application 3		Application 4	
Student	Matched-applicant group	School average SAT	School admissions decision	School average SAT	School admissions decision	School average SAT	School admissions decision	School average SAT	School admissions decision
Student A	1	1280	Reject	1226	Accept*	1215	Accept	na	na
Student B	1	1280	Reject	1226	Accept	1215	Accept*	na	na
Student C	2	1360	Accept	1310	Reject	1270	Accept*	1155	Accept
Student D	2	1355	Accept	1316	Reject	1270	Accept*	1160	Accept
Student E	2	1370	Accept*	1316	Reject	1260	Accept	1150	Accept
Student F	Excluded	1180	Accept*	na	na	na	na	na	na
Student G	Excluded	1180	Accept*	na	na	na	na	na	na
Student H	3	1360	Accept	1308	Accept*	1260	Accept	1160	Accept
Student I	3	1370	Accept*	1311	Accept	1255	Accept	1155	Accept
Student J	3	1350	Accept	1316	Accept*	1265	Accept	1155	Accept
Student K	4	1245	Reject	1217	Reject	1180	Accept*	na	na
Student L	4	1235	Reject	1209	Reject	1180	Accept*	na	na
Student M	5	1140	Accept	1055	Accept*	na	na	na	na
Student N	5	1145	Accept*	1060	Accept	na	na	na	na
Student O	No match	1370	Reject	1038	Accept*	na	na	na	na

* Denotes school attended.

na = did not report submitting application.

The data shown on this table represent hypothetical students. Students F and G would be excluded from the matched-applicant subsample because they applied to only one school (the school they attended). Student O would be excluded because no other student applied to an equivalent set of institutions.

TABLE II

MEANS AND STANDARD DEVIATIONS OF THE C&B DATA SET

	Unweighted		Weighted*			
	Full sample		Full sample		Matched applicants	
Variable	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
Log(earnings)	11.121	0.757	11.096	0.747	11.148	0.737
Annual earnings (1995 dollars)	86,768	62,504	84,219	60,841	88,276	62,598
Female	0.391	0.488	0.392	0.488	0.385	0.487
Black	0.059	0.235	0.050	0.218	0.050	0.219
Hispanic	0.016	0.124	0.013	0.115	0.014	0.117
Asian	0.027	0.162	0.023	0.150	0.027	0.163
Other race	0.003	0.059	0.003	0.059	0.003	0.057
Predicted log (parental income)	9.999	0.354	9.984	0.353	9.997	0.349
Own SAT/100	11.820	1.661	11.672	1.634	11.875	1.632
School average SAT/100	11.949	0.928	11.655	0.943	11.812	0.943
Net tuition (1976 dollars)	2733	1077	2454	1145	2651	1094
Log(net tuition)	7.781	0.591	7.647	0.622	7.749	0.582
High school top 10 percent	0.427	0.495	0.418	0.493	0.427	0.495
High school rank missing	0.360	0.480	0.356	0.479	0.355	0.478
College athlete	0.100	0.300	0.078	0.268	0.085	0.279
Average SAT/100 of schools applied to	11.678	0.928	11.513	0.940	11.601	0.991
One additional application	0.222	0.416	0.225	0.417	0.490	0.500
Two additional applications	0.230	0.421	0.214	0.410	0.366	0.482
Three additional applications	0.176	0.380	0.156	0.363	0.134	0.340
Four additional applications	0.047	0.211	0.040	0.196	0.011	0.104
Undergraduate percentile rank in class	50.703	28.473	50.791	28.267	51.666	28.268
Attained advanced degree	0.565	0.496	0.542	0.498	0.573	0.495
Graduated from college	0.846	0.361	0.839	0.367	0.862	0.345
Public college	0.282	0.450	0.413	0.492	0.329	0.470
Private college	0.540	0.498	0.442	0.497	0.523	0.500
Liberal arts college	0.178	0.382	0.145	0.353	0.148	0.355
N	14,238		14,238		6,335	

* Means are weighted to make the sample representative of the population of students at the C&B institutions.

TABLE III

LOG EARNINGS REGRESSIONS USING COLLEGE AND BEYOND SURVEY, SAMPLE OF MALE AND FEMALE FULL-TIME WORKERS

	Model					
	Basic model: no selection controls		Matched-applicant model	Alternative matched-applicant models		
Variable	Full sample	Restricted sample	Similar school-SAT matches*	Exact school-SAT matches**	*Barron's* matches***	Self-revelation model
	1	2	3	4	5	6
School-average SAT score/100	0.076	0.082	−0.016	−0.106	0.004	−0.001
	(0.016)	(0.014)	(0.022)	(0.036)	(0.016)	(0.018)
Predicted log(parental income)	0.187	0.190	0.163	0.232	0.154	0.161
	(0.024)	(0.033)	(0.033)	(0.079)	(0.028)	(0.025)
Own SAT score/100	0.018	0.006	−0.011	0.003	−0.005	0.009
	(0.006)	(0.007)	(0.007)	(0.014)	(0.005)	(0.006)
Female	−0.403	−0.410	−0.395	−0.476	−0.400	−0.396
	(0.015)	(0.018)	(0.024)	(0.049)	(0.017)	(0.014)
Black	−0.023	−0.026	−0.057	−0.028	−0.057	−0.034
	(0.035)	(0.053)	(0.053)	(0.049)	(0.039)	(0.035)
Hispanic	0.015	0.070	0.020	−0.248	0.036	0.007
	(0.052)	(0.076)	(0.099)	(0.206)	(0.066)	(0.053)
Asian	0.173	0.245	0.241	0.368	0.163	0.155
	(0.036)	(0.054)	(0.064)	(0.141)	(0.049)	(0.037)
Other/missing race	−0.188	−0.048	0.060	−0.072	−0.050	−0.192
	(0.119)	(0.143)	(0.180)	(0.083)	(0.134)	(0.116)
High school top 10 percent	0.061	0.091	0.079	0.091	0.079	0.063
	(0.018)	(0.022)	(0.026)	(0.032)	(0.024)	(0.019)
High school rank missing	0.001	0.040	0.016	0.029	0.025	−0.009
	(0.024)	(0.026)	(0.038)	(0.066)	(0.027)	(0.022)
Athlete	0.102	0.088	0.104	0.169	0.093	0.094
	(0.025)	(0.030)	(0.039)	(0.096)	(0.033)	(0.024)
Average SAT score/100 of schools applied to						0.090
						(0.013)
One additional application						0.064
						(0.011)
Two additional applications						0.074
						(0.022)
Three additional applications						0.112
						(0.028)
Four additional applications						0.085
						(0.027)
Adjusted R^2	0.107	0.110	0.112	0.142	0.106	0.113
N	14,238	6,335	6,335	2,330	9,202	14,238

Each equation also includes a constant term. Standard errors are in parentheses and are robust to correlated errors among students who attended the same institution.

Equations are estimated by WLS and are weighted to make the sample representative of the population of students at the C&B institutions.

* Applicants are matched by the average SAT score (within 25 point intervals) of each school at which they were accepted or rejected. This model includes 1,232 dummy variables representing each set of matched applicants.

** Applicants are matched by the average SAT score of each school at which they were accepted or rejected. This model includes 654 dummy variables representing each set of matched applicants.

*** Applicants are matched by the *Barron's* category of each school at which they were accepted or rejected. This model includes 350 dummy variables representing each set of matched applicants.

Welcome to the Machine

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Machine Learning meets (Mostly Harmless) Econometrics

- Machine learning (ML): explosion in computer science, key to provide good predictions.
- In this course we don't care about predictions. We care about estimating causal effects.
- However, ML methods can still be useful when pursuing estimation of causal effects.
- In particular, can ML help us automatically select the right X's for a selection on observables strategy?

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- However, ML methods can still be useful when pursuing estimation of causal effects.
- In particular, can ML help us automatically select the right X's for a selection on observables strategy?
- Next slides will provide some answers to this question while trying to explain why predictions problems are different from causal analysis.
- Heavily based on Angrist-Frandsen (2020): "Machine Labor" , another good reference is Mullainathan-Spiess (2019) in JEP.

What do we know in terms of prediction?

- Regression provides best linear approximation to the CEF by solving

$$\beta = \underset{b}{\operatorname{argmin}} \, \mathrm{E}[(Y - X_i' b)^2] = \mathrm{E}(X_i X_i')^{-1} \mathrm{E}(X_i Y_i) \tag{8}$$

- CEF is the "best" function to predict a Y_i given a $K \times 1$ vector X_i.
- Given a random sample, my LS estimator is therefore

$$\hat{\beta}_{LS} = (\frac{1}{N} \sum_{i=1}^{N} X_i X_i')^{-1} (\frac{1}{N} \sum_{i=1}^{N} X_i Y_i) \tag{9}$$

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- So why Machine Learning even exists? Things look different when we have to do out-of-sample predictions.
- Suppose I want to predict Y_{N+1} given X_{N+1} and observation $N+1$ is not included in the sample that I used to estimate $\hat{\beta}_{LS}$. Prediction based on upon least squares:

$$\hat{y}_{N+1} = X_{N+1}' \hat{\beta}_{LS} \tag{10}$$

- ML methods deploy regularization techniques (essentially to avoid over-fit). Class of estimators defined by

$$\beta_{ML} = \underset{b}{\operatorname{argmin}} \, \mathrm{E}[(Y - X_i' b)^2 + \lambda ||b||_q^q] \tag{11}$$

where λ is a regularization constant and $||b||_q = \left(\sum_k |b|^q\right)^{1/q}$

So What?

- But who cares about predictions? We want to estimate causal effects in this course.
- ML can still be useful because it allows us to see true generalizable "patterns" in the data $\rightarrow$ recover complex structure that was not specified in advance.
- Can a machine tell us what controls to use when wanting to estimate the causal effect of a particular treatment?
- Let's apply this logic in the context of Dale and Krueger (2002).

DK Research Design

- Remember DK original equation

$$Y_i = \alpha + \rho D_i + \eta_i \tag{12}$$

- DK proposed a set of variables, A_i, that can be used to condition on to invoke the CIA assumption

$$\mathrm{E}[\eta_i | D_i, A_i] = \mathrm{E}[\eta_i | A_i] \tag{13}$$

- We are not as smart as DK. However, we wonder if the machine can tell us what is the right set of control to use to derive the causal effect of D_i on Y_i.
- There is a dictionary of (potential) control variables W_i with size $m > N$. We assume that within this very large set of variables it is possible to invoke a CIA

$$\mathrm{E}(\eta_i | D_i, W_i) = \mathrm{E}(\eta_i | W_i) = \gamma_0' W_i + r_{0i} \tag{14}$$

  - r_{0i} is an approximation error.
  - Sparsity assumption: only a subset of γ_0 ($s_0 < N$) are truly different from zero (and we observe the entire dictionary).
  - We don't know which are the "useful" controls but LASSO will be able to pick them up for us under the sparsity assumption.
- Similar assumption used for the propensity score

$$\mathrm{E}(D_i | W_i) = \gamma_1' W_i + r_{1i} \tag{15}$$

Double Selection Lasso

- Belloni, Chernozhukov, and Hansen (2014): Machine learning approach to covariate selection under approximate "sparsity" (most controls irrelevant)
  1. LASSO the p-score (D_i on W_i), get relevant subset of regressors (W_{Di})
  2. LASSO the outcome (Y_i on W_i), get relevant subset of regressors (W_{Yi})
  3. Regress Y_i on $(D_i, W_{Di}, W_{Yi}) \Rightarrow$ "oracle" estimate (as good as if you knew the coefficients on the whole set of potential controls)
- Interaction controls with D_i to generalize to ATE/TNT/TOT
- Advantages: no fiddling around with which W_i's or specification tests

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- Advantages: no fiddling around with which W_i's or specification tests
- Recent application: Dube-Jacobs-Naidu-Suri(2020) to test monopsony in online job markets (AER:I, 2020).

Table 2: Post-Lasso Estimates of Elite College Effects

	Double-selection (PDS)			Outcome selection			All controls
	plugin (16)	C.V. λ	cvlasso	plugin (16)	C.V. λ	cvlasso	OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	A. Private School Effects						
Estimate	0.038	0.020	0.040	0.046	0.043	0.042	0.017
	(0.040)	(0.039)	(0.041)	(0.041)	(0.043)	(0.043)	(0.039)
No. of controls	18	100	112	10	35	50	303
	B. Effects of School Average SAT/100						
Estimate	-0.009	-0.013	-0.009	-0.008	-0.009	-0.008	-0.012
	(0.020)	(0.018)	(0.019)	(0.020)	(0.019)	(0.019)	(0.018)
No. of controls	24	151	58	10	34	43	303
	C. Effects of Attending Schools Rated Highly Competitive +						
Estimate	0.068	0.051	0.073	0.076	0.080	0.082	0.053
	(0.033)	(0.033)	(0.033)	(0.031)	(0.032)	(0.032)	(0.033)
No. of controls	17	185	106	10	34	43	303

Notes: The sample size is 14,238. Estimates in columns 1-3 are from a post-double-selection (PDS) lasso procedure. Results in columns 4-6 are from a procedure applying lasso to a reduced-form regression of the the outcome on the dictionary of controls. Columns 1 and 4 show results using the Stata 16 lasso linear command to select controls with a plug-in penalty, and OLS to compute the estimates. Columns 2 and 5 use lasso linear with 10-fold cross validation to select the penalty. Columns 3 and 6 use Stata 15 (Lassopack) cvlasso to select the penalty, rlasso to select controls, and OLS to compute estimates. See the appendix for detals. Column 7 reports OLS estimates including the entire set of controls. Controls include those used for column 5 of the previous table plus the following: indicators for being accepted to two colleges, three colleges, and four or more colleges; indicators for being rejected from one college, two colleges, three colleges, and four or more colleges; the number of schools applied to; the average SAT score among schools at which the applicant was accepted; the average SAT score among schools from which the applicant was rejected; the highest average SAT score across schools at which the applicant was accepted; the highest average SAT score across schools from which the applicant was rejected; the lowest average SAT score among schools at which the applicant was accepted; the lowest average SAT score among schools from which the applicant was rejected, and all two-way interactions of the above variables. The control dictionary contains 384 variables. OLS estimates use weights, and are reported with robust standard errors clustered by institution. All lasso commands use regressor-specific penalty loadings.

Selection on Observables: The Bottom Line

- There is obviously no theorem about when controlling for observables is sufficient to eliminate selection bias. This depends on the quality of the covariates and the nature of the selection process
- Compelling selection on observable strategies rely on good institutional knowledge and a plausible model of treatment assignment.
- DSL of Belloni et al. (2014) might be nice complement in research toolbox (as robustness)
- Even when rich covariates are available, lack of common support may limit the set of treatment effects that can be identified.